



Econometrics II

Homework 2: MLL & GMM

Due: Friday, Farvardin 31, 23:59

- Please name your file like `lastname.firstname.HW#.zip`.
 - Email your work to `unesrom@gmail.com` and `airanpour22@gmail.com` before the deadline.
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Question 1

- a) What is the mean and variance of a χ^2 random variable with d degree of freedom? Write down the two moment conditions. Based on the number of parameters and moments, justify using GMM.
- b) Use *Stata* and draw a sample of a size 500. Then use the two-step GMM to estimate the parameter using sample moment condition. (Don't forget to set the seed in *Stata* for robust results.)

Question 2

- a) Use the *mepsdocvis* dataset provided in the zip file. Consider Poisson regression of *docvis* on an intercept, *private*, *chronic*, and *income*. First explain what does Poisson command do in *Stata*. Then find the following: the sample average prediction of *docvis*; the average prediction if we use the Poisson estimates to predict *docvis* for males only; the prediction for someone who is privately insured, has a chronic condition, and has an income of \$20,000 (so *income*=20). Provide a direct interpretation of the estimated Poisson coefficients.
- b) **(extra credit)** First explain the marginal effect (ME) in nonlinear models. Then find the marginal effect of changing regressors on the conditional mean in the following ways (use predefined *Stata* commands): the AME using calculus methods for all regressors; and the MER for someone who is privately insured, has a chronic condition, and has an income of \$20,000 (so *income*=20).

Question 3

In this problem, you will use the Simulated Method of Moments (SMM) to estimate the marginal propensity to consume (MPC) in a simple economic model.

Consider a basic consumption-saving model where the consumer decides on consumption (C) and saving (S) based on their income (Y). The model is given by:

$$C = \alpha \cdot Y$$

$$S = Y - C$$

where α represents the marginal propensity to consume.

1. What distribution for Y do you think best matches the real-world data?

Simulate income data (Y) and calculate consumption (C) and saving (S) for a given value of α according to your proposed distribution.

2. Compute the empirical moments (mean and variance) of C and S .
3. Estimate the value of α that makes the simulated moments as close as possible to the empirical moments observed in real-world data. (You might need to take a guess about the real-world data moments according to the literature. Then start with a value for α and update it based on the optimization results.)
4. Discuss the economic significance of the estimated α .

Question 4

Consider the following probability distribution:

$$f(x|\theta, K) = \theta k \theta x^{-\theta-1}$$

Where $K \geq 0$ and $X_1, X_2, \dots, X_n$ is an i.i.d sample of size n .

- a) Find the Maximum Likelihood Estimator of θ .
- b) Using Method of Moments, find a formula for parameter θ .

Question 5

Consider the following probability distribution:

$$f(x|\sigma) = \frac{1}{2\sigma} \exp\left(-\frac{|x|}{\sigma}\right)$$

Use the Method of Moments to estimate the parameter σ .

Question 6

Individuals are asked at time t to remember what they had been through before. We have a model that predicts the probability of correct recall $p(w, t)$ where $W = (w_1, w_2)$ is the vector of parameters. The model is given by:

$$p(w, t) = w_1 \exp(-w_2 t)$$

We have a data containing m observations of $y_i (0 \leq y_i \leq 1)$ which represents the proportion of correct recalls at time $t_i (i = 1, 2, \dots, m)$: $y_i = \frac{x_i}{n}$, where x_i is the number of correct answers that follows a binomial distribution with probability $p(w, t)$ and n is the number of independent tests.

- a) Write the Log Likelihood function of the model.
- b) Find the first order conditions for the Maximum Likelihood Estimator.